

Segment-8

- ① Phases in selection
- ② Hardware, software selection criteria
- ③ Security, security issues, example, threat, goals
- ④ Vulnerabilities,
- ⑤ Control policy,
- ⑥ Control measures

Phases in selection: ① Hardware selection
② Software selection

Hardware selection criteria:

- ① Hardware must support current software.
- ② Hardware must be compatible with existing or planned networks.
- ③ Hardware must be upgradeable and expandable to meet the needs of the next planning interval.
- ④ Hardware warranties must be of an appropriate length.
- ⑤ Hardware maintenance must be performed by ^{remote} vendor / personnel
- ⑥ Whenever feasible, hardware standards will dictate ^{personnel} procurement of like brands and configuration. to

Simplify installation and support.

⇒ Hardware cost must be within the budget.

⇒ Hardware must be reliable.

⇒ Hardware vendor should be of high reputation.

Software criteria selection:

⇒ Software must be compatible with current and future hardware needs.

⇒ Software must be standardized throughout the business to improve purchasing power, simplify training etc.

⇒ Software must comply with current standards set by technology leadership.

⇒ Software must support and enhance business goals.

⇒ Software must be user-friendly.

⇒ Software vendor should provide user training.

⇒ Software vendor should be of high reputation.

⇒ Software vendor should provide user manuals.

Comparison of system

• Reliability - (no downtime)
- from system protection

• Exposure - (no disclosure of data (confidentiality on a piece of paper))

Security issues examples:

These

• Exposure: let the good window open
no system (system)
security (no proper door lock)
• Reliability: from power
non power

Security: security means,

Prevention - take measures that prevent your assets from being damaged.

Detection - take measures so that you can detect when, how and by whom an asset has been damaged.

Reaction - take measures so that you can recover your assets or to recover from a damage to your assets.

Security issues: ① Exposure ③ Threats
② Vulnerability ④ Control

① Exposure: A single instance when the system is open to damage.

② Vulnerability: Weakness in the system that might be exploited to cause loss or harm.

③ Threats: objects, persons, or other entities that represent a potential danger to the system.

④ Control: Protective measure - action, procedure or techniques to reduce vulnerability.

Security issues example:

House

- Vulnerability: Poor home
- Security (no proper door lock, no alarm system)
- Exposure: let the door/window open

Computer system

- Vulnerability - Poor system protection (no antivirus)
- Exposure - Unauthorized disclosure of data (write password on a piece of paper)

- Threats: Burglar / Thieves
- Control: deploy alarm system, more padlock

- Threats - Hackers, Virus
- Control - Install Firewall, install antivirus, software patch

Security Threat: security threats are potential risks that can harm computer system and organizations.

System - Collection of hardware, software, storage media & people that organization uses to performing computing task.

Asset - Organizational resource that is being protected.

There are 4 kinds of threats. They are -

- ① Interruption
- ② Interception
- ③ Modification
- ④ Fabrication

① Interruption: Asset of the system becomes lost
Example: Destruction of hardware, software, or data.

② Interception: Unauthorized access to an asset of the system.
Example: wiretapping, break-ins

③ Modification: Tampering the asset of the system by unauthorized party.
Example: change the value of the database.

④ Fabrication: Skill-full modifications of the system that could be virtually indistinguishable from real thing.

Example: Intruder might insert spurious transaction to a network communication system.

Security Goals: Prevent the vulnerability from being exploited.

3 types of security goals / 3 characteristics:

① Confidentiality: Assets are accessible only by authorised party.

② Integrity: Assets can be modified only by authorised party or only in authorised ways.

③ Availability: Assets are accessible to authorised party.

Vulnerabilities: A vulnerability is a weakness, flaw or other shortcoming in a system. 3 types of vulnerabilities-

① Hardware - It is a flaw or weakness in the design, implementation, or operation of a computer system hardware components, such as processors, memory or network devices.

Ex: Default password

② Software - Software can be replaced, changed or destroyed maliciously, modified, deleted or misplaced accidentally.

⇒ These vulnerabilities are usually exploited in software attacks.

Example - Virus, modification, Information leak etc.

③ Data vulnerability - A data vulnerability is a weakness that can be exploited by cyber criminals to gain unauthorized access to a computer system.

Example: Install spyware.

Example: Intruder might insert spyware to a network communication system.

Control policy / Method of defense:

It attempt to prevent a computing system vulnerability and preserving confidentiality, integrity and availability.

There are 4 method of defense. They are -

① Encryption

② Software controls

③ Hardware controls

④ Policies

① Encryption:

- Most powerful tool
- Provides confidentiality and integrity
- Transforming data so that it is unintelligible to the outside observer.

② Software / program Control:

- Internal program controls
- Operating system and network system
- Independent control programs
- Development controls

③ Hardware Control:

Using hardware devices in providing computer security. Example: Smart card, locks.

④ Policies:

Agreed upon procedures through legal and ethical control.

example: Frequent change of password, cyber law etc.

Control measures / 4 primary methods of system

Protection: (what makes a system secure?)

- ① System access control
- ② Data access control
- ③ system and security administration
- ④ system design

① System access Control:

- Identification and Authentication
 - ⇒ something you know (password)
 - ⇒ something you have (keys, tokens)
 - ⇒ something you are (fingerprint, handprint)
- system audit and log ⇒ keep track of what you are doing in a system.
- Protecting login and password entry
 - ⇒ Limited attempts
 - ⇒ Limited time periods
 - ⇒ Password expiration
- password storage protection
 - ⇒ Encryption
 - ⇒ Access control

② Data access Control: Monitor who can access what data and for what purpose

- Discretionary access control ⇒ How to protect asset & types of access allowed.
 - ACL → Access Control list
- Mandatory access control ⇒ system protect your asset.

③ System and Security Administration:

so It is a set of administrative security controls that include any security measures focused on managing people.

④ System design: → take advantage of hu and su security characteristics.

- ⇒ Development of the system general concept.
- Preparation of a set of working documentation.
- ⇒ Hardware specification.